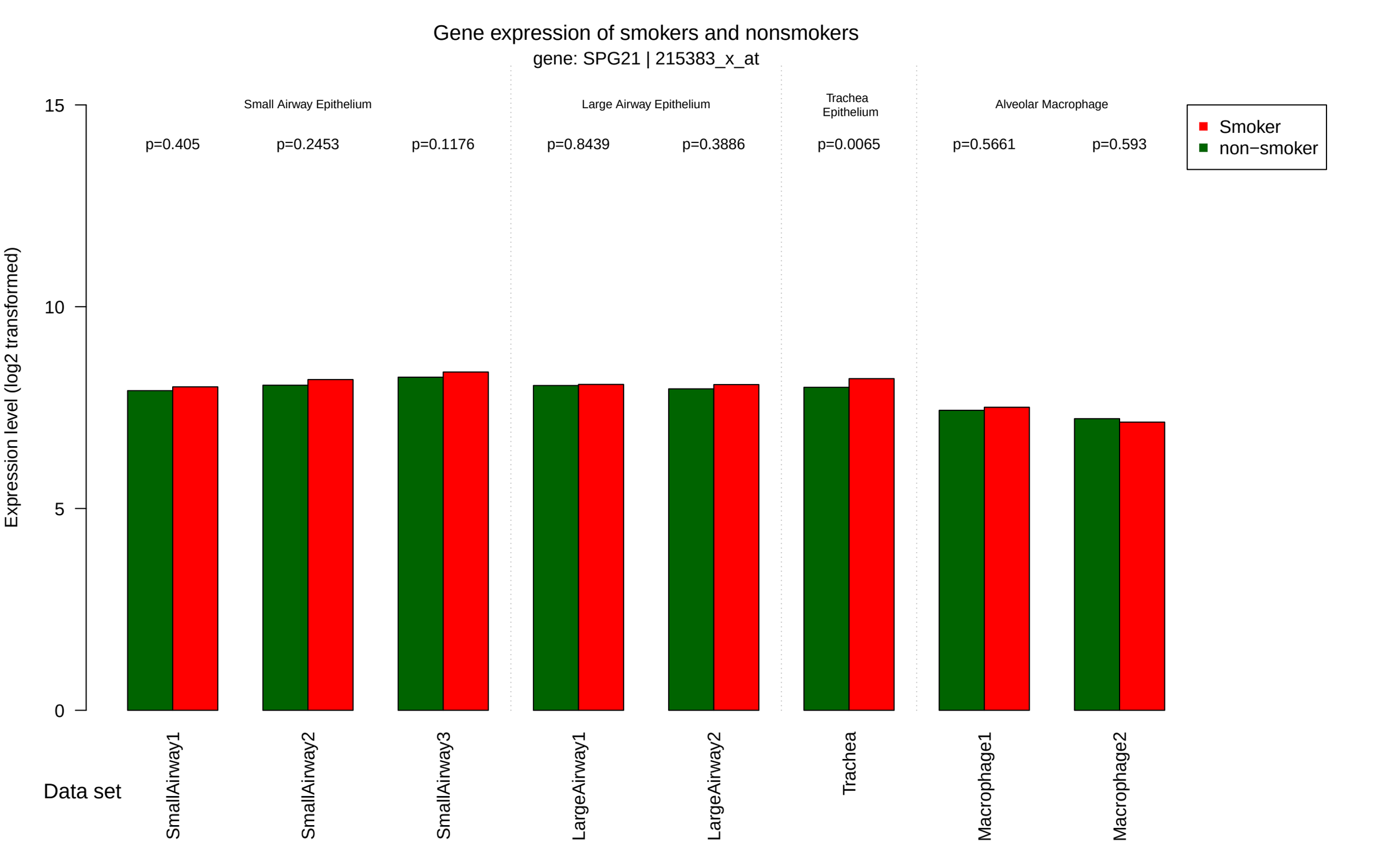
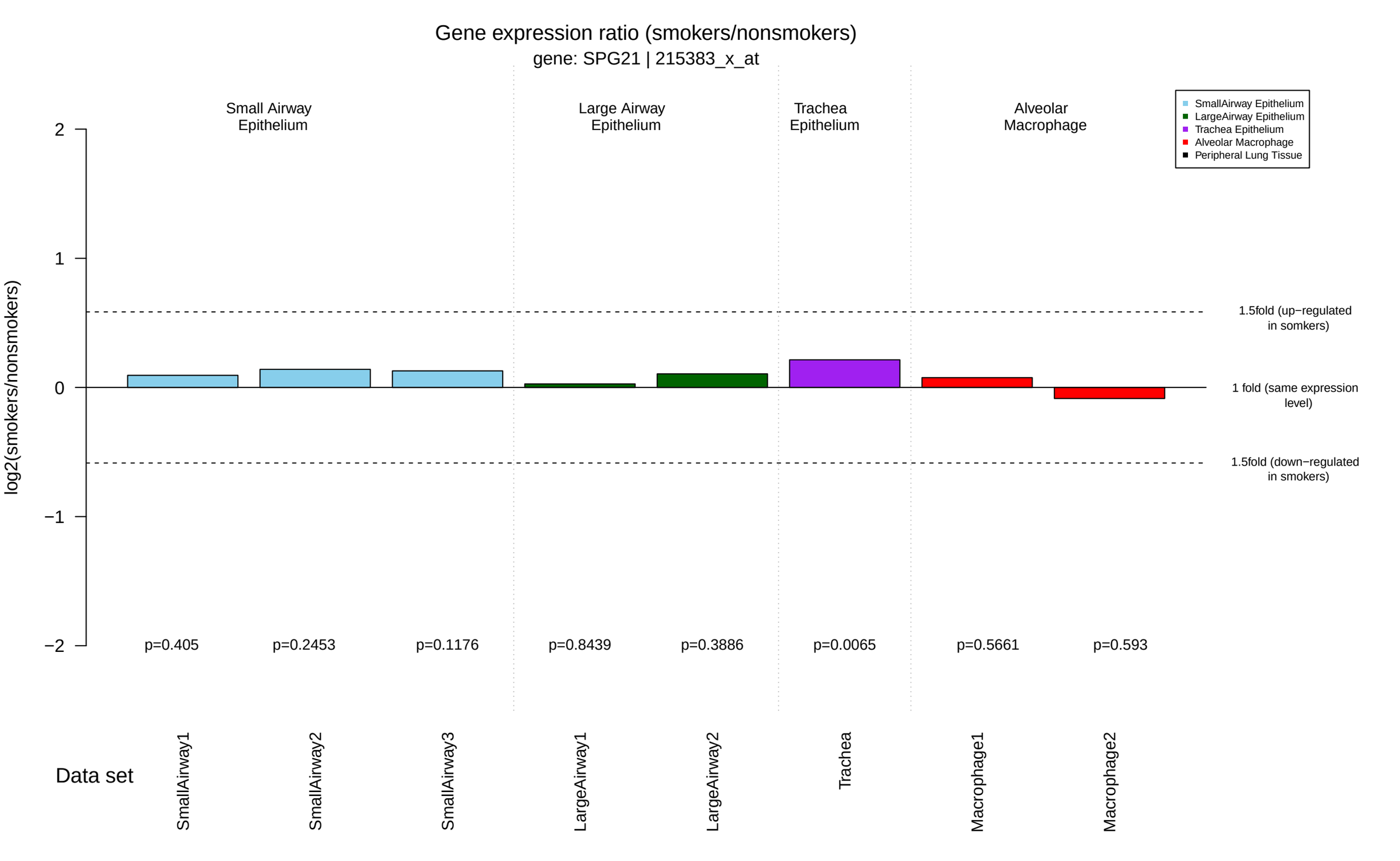
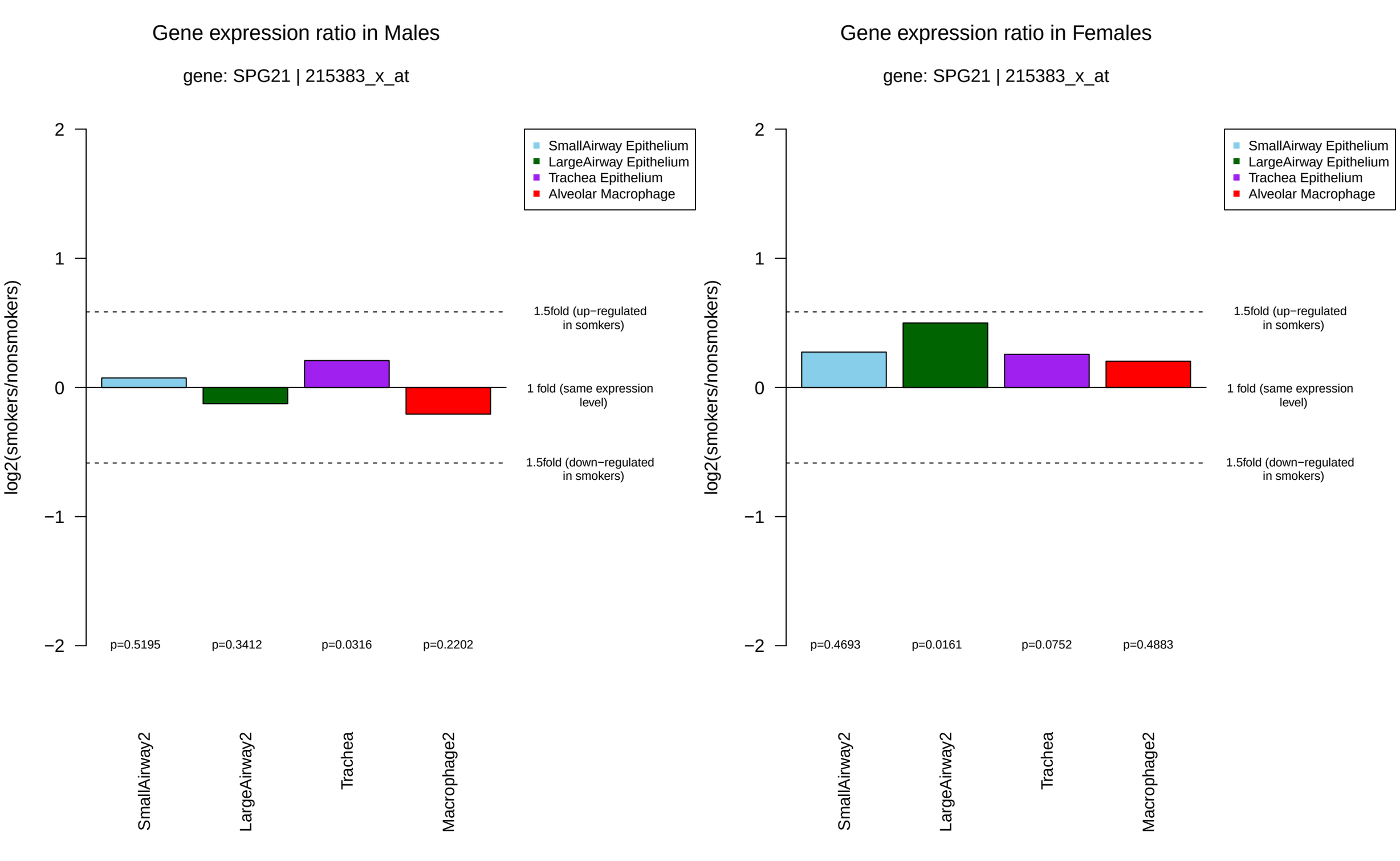
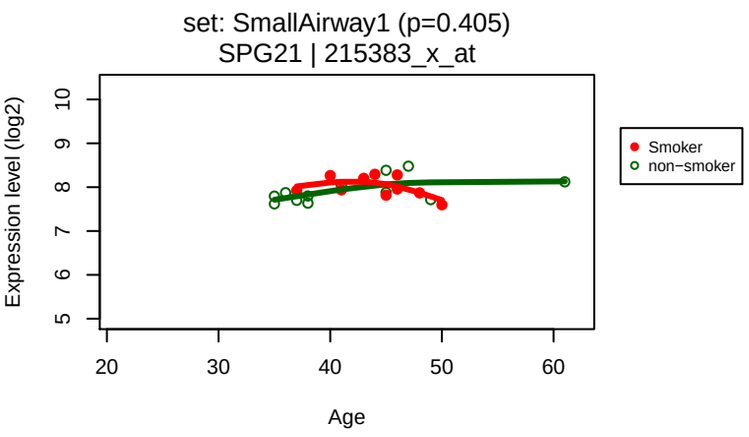
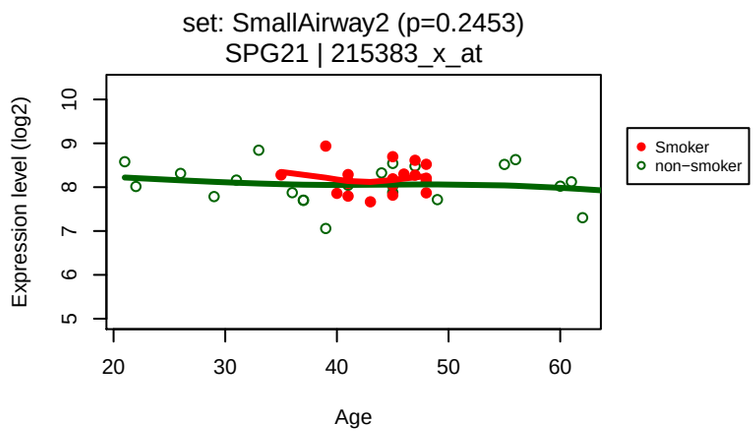
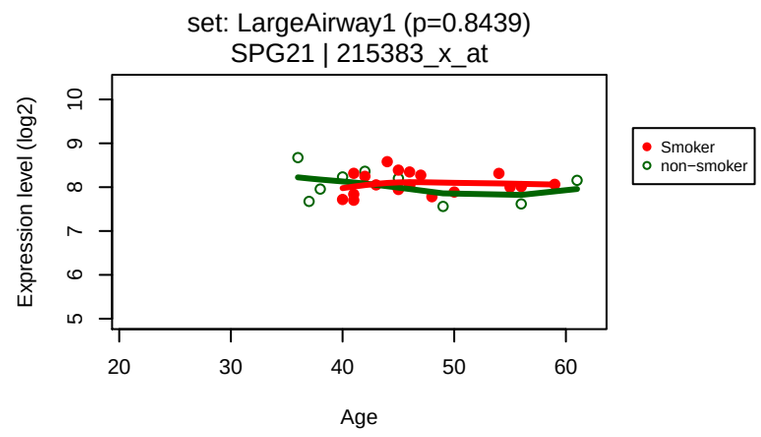

Gene expression of smokers and nonsmokers
gene: SPG21 | 215383_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.405
p=0.2453
p=0.1176
p=0.8439
p=0.3886
p=0.0065
p=0.5661
p=0.593
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

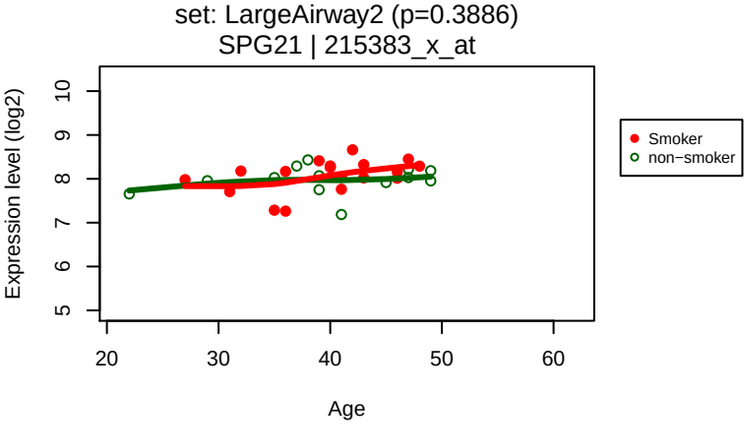

Gene expression ratio (smokers/nonsmokers)
gene: SPG21 | 215383_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.405
p=0.2453
p=0.1176
p=0.8439
p=0.3886
p=0.0065
p=0.5661
p=0.593
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

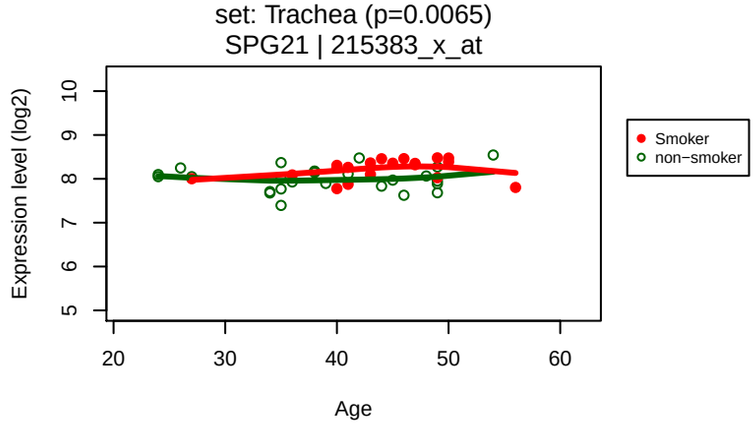

Gene expression ratio in Males
gene: SPG21 | 215383_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5195
p=0.3412
p=0.0316
p=0.2202
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPG21 | 215383_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4693
p=0.0161
p=0.0752
p=0.4883
SmallAirway2
LargeAirway2
Trachea
Macrophage2

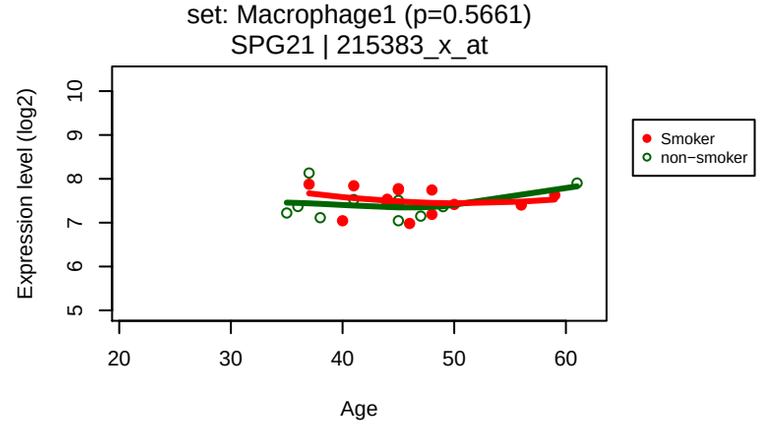

set: SmallAirway1 (p=0.405)
SPG21 | 215383_x_at
Expression level (log2)
Age
Smoker
non−smoker

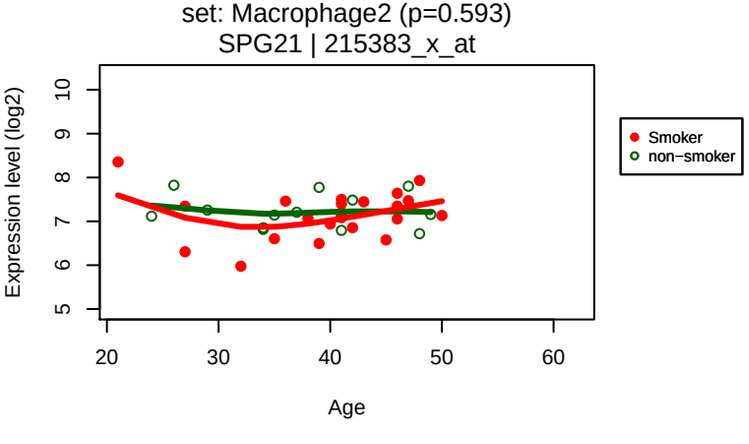

set: SmallAirway2 (p=0.2453)
SPG21 | 215383_x_at
Expression level (log2)
Age
Smoker
non−smoker

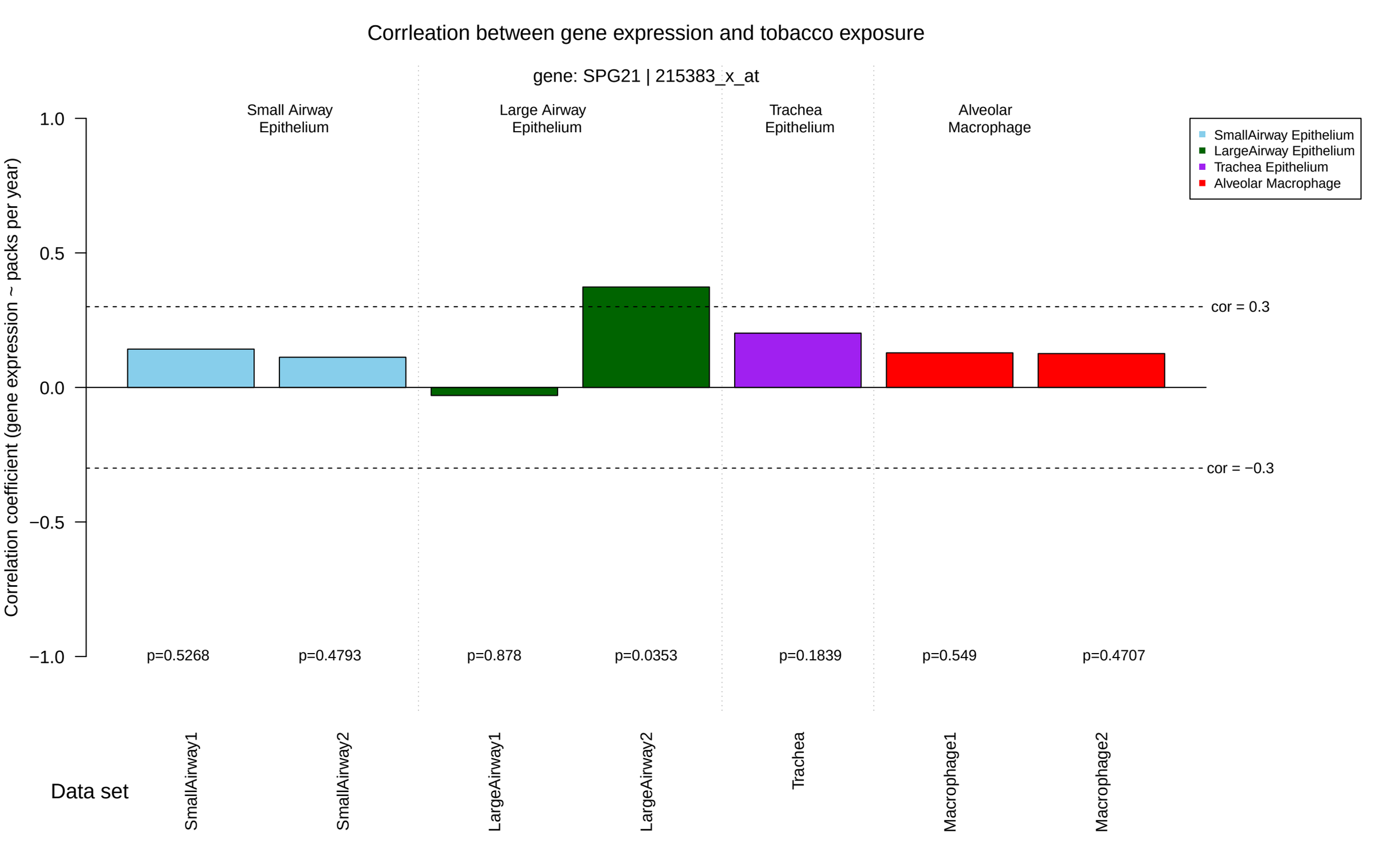

set: LargeAirway1 (p=0.8439)
SPG21 | 215383_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3886)
SPG21 | 215383_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0065)
SPG21 | 215383_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5661)
SPG21 | 215383_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.593)
SPG21 | 215383_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPG21 | 215383_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5268
p=0.4793
p=0.878
p=0.0353
p=0.1839
p=0.549
p=0.4707
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2